INTRODUCTION

Having covered yield-based duration and convexity measures, we now introduce curve-based measures of a bond's price sensitivity to changes in a benchmark yield curve and when cash flows are uncertain. We show how the change in a bond's full price is estimated by combining curve-based duration and convexity sensitivity measures, discuss uses of these approximate measures by issuers and investors, and explain their benefits and limitations. We also introduce key rate duration as a measure of interest rate risk across the term structure. Finally, we show that benchmark yield changes and credit spreads for issuers of lower credit quality are negatively correlated, especially during periods of market distress, establishing the benefit of an empirical versus analytical approach.

Category
Curve-Based and Empirical Fixed-Income Risk Measures

Learning Module Overview

- Yield duration and convexity estimates of interest rate risk are useful only for small changes in yields. Effective duration and effective convexity are valid for both small and large changes in yields.
- Effective duration and effective convexity are useful for gauging the interest rate risk of bonds whose future cash flows are uncertain.
- Effective duration and effective convexity can be used to estimate the percentage change in a bond's full price for a given shift in the benchmark yield curve.
- A key rate (or partial) duration is a measure of a bond's sensitivity to a change in the benchmark yield at a specific maturity. Key rate duration data, along with forecasted shifts in the benchmark curve, allow a portfolio to be rebalanced to improve its return.
- The sum of weighted key rate durations of the bonds in a portfolio are equal to the effective duration of the entire portfolio.
- Analytical duration and convexity are estimated duration and convexity statistics using mathematical formulas. Empirical duration and convexity are estimated using historical data that incorporate various factors affecting bond prices.
- When deciding whether to use an empirical or analytical measure, the correlation between benchmark yields and credit spreads must be considered.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of the learning module.

1. For which of the following-fixed income instruments would an investor *most likely* use yield duration and convexity to measure interest rate risk rather than effective duration and effective convexity?

A. 30-year fixed-rate bond
B. Callable bond
C. Mortgage-backed security

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A is correct. The callable bond and mortgage-backed security have embedded options, which yield duration and convexity cannot account for. The 30-year fixed-rate bond does not, so either method could be applied.

2. Calculate the effective duration of a bond given the following:

- PV_0 = 102.208.
- Price with the benchmark yield curve shifted up 25 bps: $PV+$ = 100.004.
- Price with the benchmark yield curve shifted down by 25 bps: $PV_$ = 103.891.

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Solution:

$$EffDur = \frac{(PV_-)-(PV_+)}{2 \times (\Delta Curve) \times (PV_0)}$$

$$EffDur = \frac{(103.891)-(100.004)}{2 \times (0.0025) \times (102.208)}$$

EffDur = 7.606

3. Calculate the effective convexity of a bond given the following:

- PV_0 = 102.208.
- Price with the benchmark yield curve shifted up 25 bps: $PV+$ = 100.004.
- Price with the benchmark yield curve shifted down by 25 bps: $PV_$ = 103.891.

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Solution:

$$EffCon = \frac{[(PV_-)+(PV_+)]-[2 \times (PV_0)]}{(\Delta Curve)^2 \times (PV_0)}$$

$$EffCon = \frac{[(103.891)+(100.004)]-[2 \times (102.208)]}{(0.0025)^2 \times (102.208)}$$

EffCon = −815.592

4. If the benchmark yield curve downward shifted by 50 bps, what would be the percentage change in the full price of a bond if its effective duration is 6.094 and its effective convexity is –230.097?

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Solution:

$$\%\Delta PV^{Full} \approx (-EffDur \times \Delta Curve) + \left[\frac{1}{2} \times EffCon \times (\Delta Curve)^2\right]$$

$$\%\Delta PV^{Full} \approx (-6.094 \times 0.005) + \left[\frac{1}{2} \times -230.097 \times (0.005)^2\right]$$

$\%\Delta PV^{Full} \approx 3.33\%$

5. The table below contains forecasts of expected changes in the benchmark yield curve by tenor.

Maturity	1 year	5 years	10 years	20 years	30 years
Curve shift at maturity	+100 bps	+150 bps	+200 bps	+250 bps	+300 bps

Your portfolio contains equally sized positions in Bond A and Bond B, which have the following key rate durations.

Bond A	5 years	1.706
Bond B	10 years	3.195

Given the information and expected changes in yields, identify the bond that could be sold to improve returns.

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Solution:

Bond B, because its expected price decline from the expected change in benchmark yields is more than twice that of Bond A.

For Bond A, $\Delta PV/PV = -1.706 \times 0.015 = -2.56\%$.

For Bond B, $\Delta PV/PV = -3.195 \times 0.020 = -6.39\%$.

6. Explain why yield duration and convexity are less useful for bonds with contingency features, such as callable bonds.

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Solution:

Yield duration and convexity assume cash flows are received on scheduled dates. Cash flows from bonds with contingency features may *not* be received on scheduled dates—if, for example, a callable bond is called by the issuer following a decrease in interest rates. Therefore, yield duration and convexity would not correctly measure the price sensitivity to interest rates.

7. When comparing analytical duration versus empirical duration, which of the following statements is correct?

A. Analytical duration and convexity are estimated duration and convexity statistics using mathematical formulas.
B. Empirical duration and convexity are estimated using historical data in non-statistical models that incorporate various factors affecting bond prices.
C. Both A and B

Hide Solution

Solution:

A is correct. Analytical duration and convexity are estimated duration and convexity statistics using mathematical formulas.

B is incorrect because statistical models are primarily used to determine empirical duration and convexity.

C is incorrect because B is incorrect.

8. True or false: Analytical duration is superior to empirical duration for bonds with credit risk.

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Solution:

False. If government bond yields are driven lower in a market stress scenario, the “flight to safety” will also cause credit spreads to widen because of an increase in expected default risk. Since credit spreads and benchmark yields are negatively correlated under this scenario, wider credit spreads will partially or fully offset the decline in government benchmark yields, resulting in lower empirical duration estimates than analytical duration estimates. Therefore, empirical duration estimates are a more accurate method of forecasting bond price changes for bonds with credit risk.

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PK Q-4 why would u take postive DeltaY as 0.5%in formula where benchmark yield curve downward shifted by 50 bps so real ans coming is 2.76%.

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